Assignment 5 Solution

```
install.packages("marginaleffects")
```

```
Installing package into '/usr/local/lib/R/site-library'
(as 'lib' is unspecified)
```

```
library(tidyverse)
```

```
── Attaching core tidyverse packages ──────────────────────── tidyverse 2.0.0 ──
✔ dplyr     1.1.4     ✔ readr     2.1.5
✔ forcats   1.0.0     ✔ stringr   1.5.1
✔ ggplot2   3.5.0     ✔ tibble    3.2.1
✔ lubridate 1.9.3     ✔ tidyr     1.3.1
✔ purrr     1.0.2
── Conflicts ────────────────────────────────────────── tidyverse_conflicts() ──
✖ dplyr::filter() masks stats::filter()
✖ dplyr::lag()    masks stats::lag()
ℹ Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```
library(marginaleffects)
```

```
my_url <- "http://datafiles.ritsokiguess.site/teak.csv"
teak <- read_csv(my_url)
```

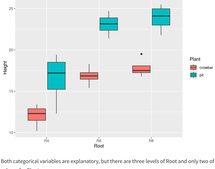
```
Rows: 24 Columns: 4
── Column specification ────────────────────────────────────────────────────────
Delimiter: ","
chr (3): Block, Plant, Root
dbl (1): Height

ℹ Use `spec()` to retrieve the full column specification for this data.
ℹ Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```
teak
```

```
# A tibble: 24 × 4
   Block Plant  Root  Height
   <chr> <chr>  <chr>  <dbl>
 1 I     pit    R4     18.2
 2 I     pit    R6     23.6
 3 I     pit    R8     24.0
 4 I     crowbar R4    12.7
 5 I     crowbar R6    15.4
 6 I     crowbar R8    16.0
 7 II    pit    R4     16.2
 8 II    pit    R6     22.7
 9 II    pit    R8     23.4
10 II    crowbar R4    11.9
# ℹ 14 more rows
```

```
ggplot(teak, aes(x = Root, y = Height, fill = Plant)) + geom_boxplot()
```

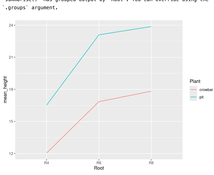


Both categorical variables are explanatory, but there are three levels of Root and only two of Plant , so its better to put Root on the x-axis and have only two colours for Plant .

Based on the boxplot when the planting method is Pit, the plant has a greater height on average than when the planting method is Crowbar, and the amount by which its higher is about the same for each root length. Thus, it is expect to see a Plant-effect that is the same for each root length that is, no interaction between planting method and root length. To reaffirm our point we can draw an interaction plot and see whether it tells the same story as the boxplots. Note (it should, except for the fact that an interaction plot uses means and the boxplots use medians)

```
teak %>%
  group_by(Root, Plant) %>%
  summarize(mean_height = mean(Height)) %>%
  ggplot(aes(x = Root, y = mean_height, colour = Plant, group = Plant)) +
geom_line()
```

```
`summarise()` has grouped output by 'Root'. You can override using the
`.groups` argument.
```



Indeed those two traces do indeed look pretty close to parallel.

```
teak.1 <- aov(Height ~ Root + Plant, data = teak)
summary(teak.1)
```

```
            Df Sum Sq Mean Sq F value   Pr(>F)
Root         2 203.15  101.57  32.131 1.15e-06 ***
Plant        1 187.60  187.60  59.344 4.18e-07 ***
Root:Plant   2   3.78    1.89   0.598    0.56
Residuals   18  56.90    3.16
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

With a P-value of 0.56, this is nowhere near significant.

```
teak.2 <- aov(Height ~ Root + Plant, data = teak)
summary(teak.2)
```

```
            Df Sum Sq Mean Sq F value   Pr(>F)
Root         2 203.15  101.57   33.48 4.14e-07 ***
Plant        1 187.60  187.60   61.83 1.52e-07 ***
Residuals   20  60.68    3.03
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

We can conclude that there are significant main effects, of root length (P-value 4.1 × 10−7) and of planting method (P-value 1.5 × 10−7). To find out what kind of effects they are, we run Tukey (which is necessary because there are three different root lengths):

```
TukeyHSD(teak.2)
```

```
  Tukey multiple comparisons of means
    95% family-wise confidence level

Fit: aov(formula = Height ~ Root + Plant, data = teak)

$Root
     diff     lwr     upr     p adj
R6-R4 5.6875 3.48403 7.89097 0.0000007
R8-R4 6.5625  4.35903 8.76597 0.0000000
R8-R6 0.8750 -1.32847 3.07847 0.5825209

$Plant
                diff     lwr     upr p adj
pit-crowbar 5.591667 4.180292 7.075041 2e-07
```

The Plant analysis here doesn't tell us anything we haven't already seen: when the Pit planting method is used, the height is on average greater than for the Crowbar method. The Root analysis, however, does tell us something new: the height for plants that had root length 4 inches is significantly less than for plants with either of the other two root lengths (which are not significantly different). There is no value in doing simple effects here, because the interaction is not significant.

```
my_url <- "http://datafiles.ritsokiguess.site/rhiz-clover.csv"
clover <- read_csv(my_url)
```

```
Rows: 60 Columns: 5
── Column specification ────────────────────────────────────────────────────────
Delimiter: ","
chr (2): comb, strain
dbl (3): weight, nitro, Npg

ℹ Use `spec()` to retrieve the full column specification for this data.
ℹ Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```
clover
```

```
# A tibble: 60 × 5
   comb   strain weight nitro  Npg
   <chr>  <chr>  <dbl> <dbl> <dbl>
 1 clover 3DOK1  1.02  31.2  30.6
 2 clover 3DOK1  0.65  23.6  36.3
 3 clover 3DOK1  0.55  14    25.5
 4 clover 3DOK1  0.86  23.1  26.9
 5 clover 3DOK1  0.9   23.4  26
 6 clover 3DOK5  1.18  59    50
 7 clover 3DOK5  1.3   36.7  28.2
 8 clover 3DOK5  0.86  30.7  45
 9 clover 3DOK5  1.62  47.3  29.2
10 clover 3DOK5  0.9   26.1  29
# ℹ 50 more rows
```

There are correctly 12 × 5 = 60 observations, and the variables of interest are as described. (The value of Npg is weight divided by nitro , which is why it has a lot of decimals.)

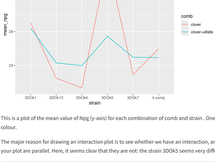
```
clover %>%
  group_by(comb, strain) %>%
  summarize(mean_npg = mean(Npg)) -> clover_means
```

```
`summarise()` has grouped output by 'comb'. You can override using the
`.groups` argument.
```

```
clover_means
```

```
# A tibble: 12 × 3
# Groups:   comb [2]
   comb          strain mean_npg
   <chr>         <chr>     <dbl>
 1 clover        3DOK1      29.8
 2 clover        3DOK13     22.5
 3 clover        3DOK4      21.4
 4 clover        3DOK5      36.3
 5 clover        3DOK7      22.9
 6 clover        s.comp     26.8
 7 clover+alfalfa 3DOK1    28.4
 8 clover+alfalfa 3DOK13   24.3
 9 clover+alfalfa 3DOK4    24.8
10 clover+alfalfa 3DOK5    27.4
11 clover+alfalfa 3DOK7    25.8
12 clover+alfalfa s.comp   24.9
```

```
ggplot(clover_means, aes(x = strain, y = mean_npg,
                         colour = comb, group = comb)) + geom_line()
```



This is a plot of the mean value of Npg (y-axis) for each combination of comb and strain . One of the explanatory variables goes on the x-axis, and the other is colour.

The major reason for drawing an interaction plot is to see whether we have an interaction, and that's assessed by making a call about whether the lines on your plot are parallel. Here, it seems clear that they are not: the strain 3DOK5 seems very different from the others. So we would expect to see an interaction.

```
clover.1 <- aov(Npg ~ comb + strain, data = clover)
summary(clover.1)
```

```
            Df Sum Sq Mean Sq F value   Pr(>F)
comb         1   6.9    6.88   0.531 0.46955
strain       5 642.3  128.45   9.916 1.47e-06 ***
comb:strain  5 228.2   45.65   3.524 0.00857 **
Residuals   48 621.8   12.95
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

The interaction is significant (which is our finding from the ANOVA), so a suitable technique would be to use simple effects to understand what that significant interaction actually means.

```
clover %>%
  filter(strain == "3DOk7") -> d1
d1.1 <- aov(Npg ~ comb, data = d1)
summary(d1.1)
```

```
            Df Sum Sq Mean Sq F value Pr(>F)
comb        1 10.322  10.322  10.51 0.0118 *
Residuals   8  7.854   0.982
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

There is a significant difference in nitrogen content between the two cultures of 3DOk7. To see which way around they are, look back at the interaction plot to see that the nitrogen content of the pure clover culture is lower than for the clover-plus alfalfa culture. Or, we can perform Tukey (which is otherwise pointless) to see the same thing:

```
TukeyHSD(d1.1)
```

```
  Tukey multiple comparisons of means
    95% family-wise confidence level

Fit: aov(formula = Npg ~ comb, data = d1)

$comb
                diff     lwr     upr     p adj
clover+alfalfa-clover 2.013193 0.586832 3.477845 0.0118345
```

The mean nitrogen content for clover plus alfalfa is higher than for just clover (by between 0.5 and 3.5 units). Tukey is otherwise pointless here because there are only two levels of comb (two cultures of this strain), and we know they are significantly different.

```
clover %>%
  filter(strain == "3DOk1") -> d2
d2.1 <- aov(Npg ~ comb, data = d2)
summary(d2.1)
```

```
            Df Sum Sq Mean Sq F value Pr(>F)
comb        1   1.01   1.013   0.09  0.772
Residuals   8  90.12  11.265
```

This is not significant

The ANOVA earlier found a significant interaction. This means that the effect of comb will be different for the different strains. In particular, we found from the simple effects analyses that there is no effect of comb for strain 3DOk1 , but there is an effect for strain 3DOk7 : that is, the effect of comb depends on which strain you are looking at.